Put more formally, using the assumptions given above, for a defect i with a probability of failure per program test of λ_i, the probability of observing a failure after T prior tests is:

$$P(T_i = T+1 \mid \lambda_i) = \lambda_i (1-\lambda_i)^T \quad (1)$$

where T_i is the test where the defect is detected (and removed).

Differentiating with respect to λ_i, it can be shown that the maximum failure probability after T tests occurs when:

$$\lambda_i = \frac{1}{T+1}$$

Substituting back into (1) and rearranging, the upper bound on the probability of failure after T tests is:

$$P(T_i = T+1) \le \frac{1}{T}\left(1-\frac{1}{T+1}\right)^{T+1} \quad (2)$$

There is a standard result that $\left(1+\frac{x}{n}\right)^n \to e^x$ as $n \to \infty$.

As the e^x asymptote is an upper bound, it follows that, for all T:

$$P(T_i = T+1) < \frac{e^{-1}}{T}$$

If we define $\theta_i(T)$ as $P(T_i=T+1)$, the probability of failure per test after T tests, then:

$$\theta_i(T) < \frac{1}{eT} \quad (3)$$

This bound is *independent of the defect failure probability*. Figure 2 below illustrates this independence for different values of λ_i.

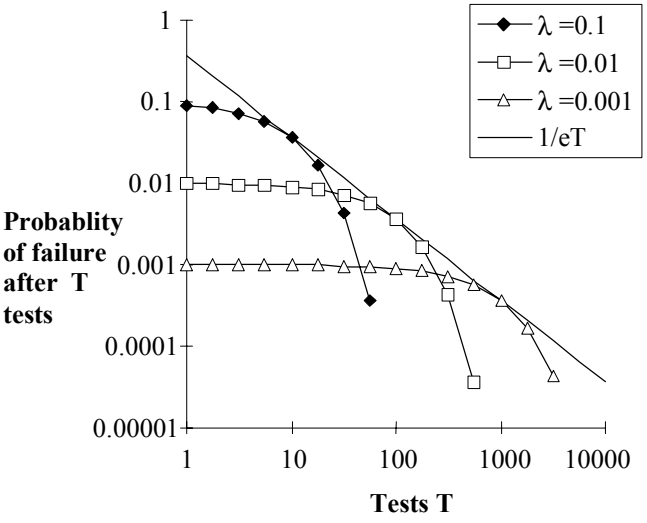


Figure 2. Illustration of the worst case bound

It is clear that, regardless of the value of λ_i, the probability of failure per test after T tests, $\theta_i(T)$, is bounded by $1/eT$.

In the worst case, the failures of all faults are disjoint, so we can sum the bounds for all N faults to derive a worst case bound for the probability of program failure after T tests, $\theta(T)$, i.e.:

$$\theta(T) < \frac{N}{eT} \quad (4)$$

This is the discrete equivalent of the continuous time bound equation in [2] where the bound on failures per unit time after a test time t is shown to be exactly N/et (or alternatively the bound on the MTTF after a test time of t, MTTF(t), is exactly et/N).

The result is surprising because it permits long-term reliability predictions to be made at t=0. If the model assumptions apply and we can estimate the number of faults N at the time of release (e.g. using estimation methods such as [3], [4], [7], [9]) the reliability growth can be bounded at *any time in the future*. Note that the theory does not tell us when (or even if) the faults will be found, but it does set a quantitative bound on the probability of program failure after testing and this bound always decreases with increasing tests (or operating time).

Figure 3 shows the application of the theory to the field reliability data of a commercial teleswitch [5]. The reliability bound is based on an estimate for N that is 50% more than the number of detected faults reported in [5]. The bars represent the measurement uncertainty in taking the average of several times to failure to estimate the current MTTF.

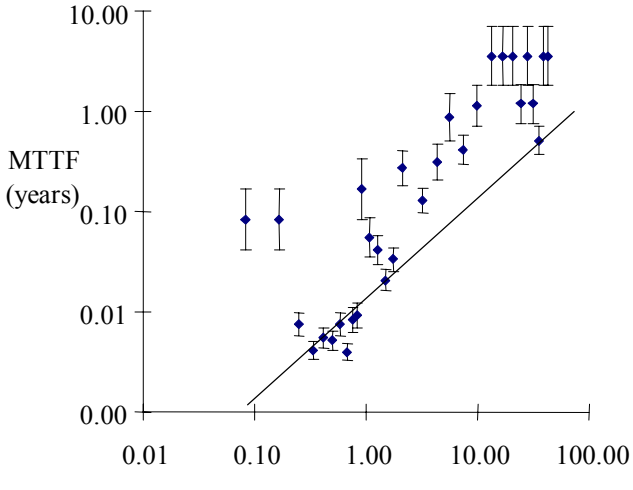


Figure 3. Reliability growth of a commercial teleswitch

The effect of violating theory assumptions (a) and (c) were examined in [2]. Both correction-induced faults and variations in the operational profile can alter the bound and cause a "saw-tooth" pattern of failure rates. It was however also shown that the worst bound should still hold over time periods that are longer than the short term fluctuations caused by changes in operational profile.

However the theory cannot predict what the reliability bound will be after a major change in the input profile (e.g. from development testing to operation). For example, in figure 3 above, there is a sudden drop in reliability after 0.1 years of usage. This might be due to a transition from field trials to full operational use—a point where there could be a dramatic change in the operational profile and where the standard reliability bound